

MEDICATION BASED; ERECTILE DYSFUNCTION

JAVEID IQBAL, A.Q. SIKANDER RIAZ*, ISHRAT BUKHARI,
MOHAMMAD JAMSHID**, SAJID BASHIR, NAWAZISH-E-HUSSAN**
AND BASHIR AHMAD**

Department of Pharmacy, University of Balochistan, Quetta

**Institute of Public Health, Bolan Medical College, Quetta*

***Faculty of Pharmacy University of the Punjab, Lahore*

ABSTRACT

Objective; the medication considered to be important for the treatment of some vital diseases and may cause erectile dysfunction. This article aim to describe some adverse drugs reaction that directly or indirectly exert its effect on the various routs of excretion process. A computer as well as manual research was made to collect the relevant data published in various journals on the drugs. The study of these drugs show positive effect on sexual performance due to the side effect of a range of commonly prescribe drugs and is important cause of noncompliance with antihypertensive treatment. This article review the drugs implicated in erectile dysfunction ED and provide advice and how to avoid it.

INTRODUCTION

Man survives earthquakes, experiences the horrors of illness, and tortures of the soul. But the most vital tragedy of all time is and will be tragedy of the bedroom (Leo Tolstoy).

Impotence, or eractile dysfunction (ED) is the inability to attain an erection adequate for satisfactory sexual intercourse. Medication is not common contributing cause of male ED (Brock., *et al.*, 1993 and Horowitz *et al.*, 1979). ED is age related (Feldman *et al.*, 1994) and as the older man is also much more likely to be receiving a Variety of drugs, a review of their medication can potentially result in the return of normal sexual function (Bansal *et al.*, 1988 and Lancet, 1981).

Drugs controlling high blood pressure:

This group of drugs most commonly concerned with ED. The action by which these anti hypertensive agent causes the ED is not well established because combination of drugs is frequently used and the incidence of ED is higher in treated hypertensive men compared to normal individual.

Beta-blockers

Association between beta-blockers and ED have been frequently reported (Brock et at., 1993 and Horowitz et at., 1979). And the incidences with propranolol quoted as upto 15%.

It has been shown that the beta-blockers, e.g. atenolol and nodolol are associated with a lower rate of ED, with obvious implication for prescribing.

The mechanisms by witch by beta-blockers lead to ED are unclear however, it has been shown that they can reduce testosterone levels, this effect is seen maximum with propranolol.

Many hypertensive patients will atherosclerotic stenosis of the pelvic vessels and therefore a reduced penile blood flow beta-blockers will further reduce this flow by lowering the systolic blood pressure.

Interestingly, propranolol has been linked to Peyronie's disease and recognized cause of ED as well as of deformity (Chang *at al.*, 1991).

Centrally acting antihypertensive

Methyldopa (aldomet) and clonidine act centrally to inhibit sympathetic out flow and therefore reduces blood pressure, both have been associated with ED.

Methyldopa has frequently been reported as a factor in ED (Hogan *et al.*, 1981). The mechanism is likely to be related to reduce libido. Methyldopa can cause an elevation in prolactin level that may lead to depressed libido and ED. This sexual function seems to be a reversible after cessation of the drug.

Erectile dysfunction with clonidine is reported (Hogan *et al.*, 1981). With a maximum incidence of 24%. As with methyldopa, the mechanism is not clear.

Diuretic

A placebo-controlled trial (Chang *et al.*, 1991), indicated a sine cause increase in ED in the thiazide diuretic treated group.

The mechanism by which thiazide diuretic lead to ED is unknown but hypothesis is to reduce intravascular volume or hypokalaemia. It has been shown (Soyka *et al.*, 1987). That thiazide have a direct inhibitory effect on relaxation of the smooth muscle in the corpora cavernosum, therefore preventing erection.

Spironolacton is an aldosterone antagonist and act as a potassium sparing diuretics, there are many reports of its association with gynaecomastia, reduced libido and ED with an incidence of between 4 to 30% (Brock *et al.*, 1993).

Psychotropic drugs

Antipsychotic

These drugs are commonly associated with sexual dysfunction (Segraves *et al.*, 1998). Including erectile problem. They have an anticholinergic effect and therefore with the parasympathetic nerve supply to the penis. These antidopaminergic action on the hypothalamus lead to increase secretion of prolactin and thus are reduced level of sex

steroids and libido.

The major tranquilizers such as the phenothiazines and butyro-phenones have a sedative effect, further reducing libido. The side effect on sexual and erectile function are thought to occur within 24 hours of treatment and resolve after therapy (Hogan *et al.*, 1981).

Antidepressant

The two main groups associated with ED are the tricyclics, e.g., imipramine and amitriptyline, and MAOIs e.g., phenazine.

These drugs act both centrally and peripherally. They have a central sedative action, reducing libido (Soyka *et al.*, 1987).

Anxiolytic

Benzodiazepine have a sedative action that reduces libido and can therefore lead to ED. The effect is more marked they are part of combination therapy with lithium. However the situation is complicated, as many of these patients will be suffering from depression or anxiety state that have a direct psychological impact on their potency (Segraves *et al.*, 1988).

Hormonal agents

Antiandrogen Drugs, e.g., cyproterone acetate and flutamide (drogenil) and leuteinising hormone releasing hormone (LHRH) analogues that leads to medical castration, e.g., gosereline and leuporelin are being used increasingly in the treatment of hormone sensitive prostate cancer, they have been strongly linked with ED as a major side effect.

Other drugs

Cimetidine

This drug has an antiandrogen action; with associated gynaecomastia and ED. Incidence of ED is lower with ranitidine.

Digoxin

This drug has been reported to have antiandrogen and estrogenic action with reduced libido and ED in long term use (Brock

et al., 1993) Gynaecomastia has also been reported.

Nicotine

When a cigarette is smoked the nicotine 'not only effect the smooth' muscle of the vascular walls leading to vasopression but also increases the tone of the erectile tissue in the corpora Cavernosum, thereby reducing potency (Lancet, 1981).

Long term nicotine abuse is associated with a atherosclerosis of internal pudendal arteries involve in blood supply to the penis, and it has been shown that smokers are twice as likely to be impotent as age-matched controls (Hogan et al., 1981).

Alcohol

Acute intoxication will lead to sedative and transient ED. Long term use leads to liver cirrhosis and associated abnormalities of the steroids metabolism. Estrogen excess and reduced testosterone level leads to decreased libido and associated ED (Feldman et al., 1994).

CONCLUSIONS

There is an increasingly awareness of sexual dysfunction in men and the associated morbidity for a man and his partner.

Sexual dysfunction is a common side effect of variety of drugs. Sexual dysfunction has been reported as the most common reason for non-compliance in-patient on antihypertensive treatment (Bansal et al., 1988 and Lancet 1981). A review of medication and informed prescribing can lead not only to a reduction of morbidity associated with sexual dysfunction, but also an increase in compliance and therefore in effectiveness of treatment for common condition such as hypertension.

If the sexual dysfunction is not related to the medication or if the medication can not be changed, then after proper assessment a choice of effective treatment option can. now be

offered.

We conclude and recommend that a sexual history including a reviews of medication, be taken in all patients at risk because a problem can not be correlated until it is discovered.

REFERENCES

- Bansal S. (1988). Sexual dysfunction in hypertensive men, a critical review. *Hypertension*. **12**: 1-10.
- Brock G.B. and Lue T.F. (1993). Drug induced male sexual dysfunction. *Drug*. **8**(6): 414-426.
- Chang S.W. et al. (1991). The impact of diuretic therapy on reported sexual function. *Archives of internal medicine*. **151**: 2402-2408; *Drug therapy*. (1987), **11**: 60-76.
- Feldman H.A. et al. (1981). Impotence and its medical and psychosocial correlates; Hogan M.J. et al. (1981). Antihypertensive therapy and male sexual dysfunction. *Psychosomatic*. **21**: 234- 237.
- Horowitz J.D. (1979) and Globel A.J. (1979). Drugs and impaired male sexual function. *Drugs*. **18**: 2.
- Medical Research Council working party on mild to moderate hypertension adverse reaction to bendrofluzide and propranolol for the treatment of mild hypertension (1981). *Lancet*. **2**: 539-543.
- Result of the Massachusetts male aging study (1994). *Jurol*. **151**: 54-61.
- Segraves R.T. (1998). Sexual side effect of psychiatric drugs. *International Journal of Psychiatric in Medicine*. **18**: 243-253.
- Soyka L.F. and Mattison D.R. Prescription drug that effect male sexual function.

